

PROJECT REPORT

STUDENT MARKS PREDICTION USING MACHINE LEARNING

Course: Fundamentals of Artificial Intelligence and Machine Learning

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1. ABSTRACT

Student performance is an important factor in education. The purpose of this project is to develop a simple Machine Learning system that predicts a student's final marks based on their study hours and previous marks.

The project uses a dataset containing study hours, previous marks, and final marks. The Linear Regression algorithm is used to learn the relationship between these values. The dataset is divided into training and testing data. The model is trained using the training data and evaluated using the R^2 score.

After training, the system allows the user to enter study hours and previous marks. The trained model then predicts the student's expected final marks.

This project demonstrates a basic and practical application of Machine Learning using Python.

2. INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every situation.

In the education field, Machine Learning can be used to analyze student-related data and predict academic performance.

In this project, a Student Marks Prediction system is developed using Python. The system uses two input features:

- Study Hours

- Previous Marks

The target value is:

- Final Marks

A Linear Regression model is trained using existing student data. After training, the model predicts the final marks of a new student based on the given input values.

The project is designed to be simple, easy to understand, and executable through the command line.

3. PROBLEM STATEMENT

Students' academic performance can depend on several factors, including their study habits and previous academic performance.

The problem addressed in this project is:

"To develop a simple Machine Learning model that predicts a student's final marks based on study hours and previous marks."

The system takes the required student information as input and provides the predicted final marks as output.

4. OBJECTIVES

The main objectives of this project are:

1. To understand the basic concept of Machine Learning.
 2. To create a simple dataset for student performance.
 3. To use study hours and previous marks as input features.
 4. To train a Linear Regression Machine Learning model.
 5. To divide the dataset into training and testing data.
 6. To evaluate the model using the R^2 score.
 7. To predict the final marks of a new student.
 8. To develop a project that can be executed through the command line.
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5. SCOPE OF THE PROJECT

The project focuses on basic student marks prediction using Machine Learning.

The current system uses:

- Study Hours
- Previous Marks

The system can be extended in the future by adding more factors such as:

- Attendance
- Assignment performance
- Internal examination marks
- Number of subjects
- Study habits

The project can therefore serve as a basic foundation for a more advanced student performance analysis system.

6. TECHNOLOGIES USED

The following technologies and tools are used:

Programming Language

Python

Python is used to implement the complete project because it is simple and provides useful Machine Learning libraries.

Libraries

Pandas

Pandas is used to read and process the CSV dataset.

Scikit-learn

Scikit-learn is used to create, train, test, and evaluate the Machine Learning model.

Machine Learning Algorithm

Linear Regression

Linear Regression is used to predict the final marks.

Development Environment

Visual Studio Code

VS Code is used to write and execute the Python program.

Version Control

Git and GitHub

Git is used for version control and GitHub is used to store and share the project repository.

7. DATASET

The project uses a simple CSV dataset named data.csv.

The dataset contains three columns:

Attribute	Description
StudyHours	Number of hours studied by the student
PreviousMarks	Marks obtained previously by the student
FinalMarks	Final marks obtained by the student

The dataset contains sample student records that are used to train and test the Machine Learning model.

Example Dataset

Study Hours	Previous Marks	Final Marks
1	45	48
2	50	52
2.5	55	57
3	60	61
4	65	67
5	70	73
6	75	77
7	80	82
8	85	87
9	90	92
10	92	95

The dataset is stored in the project repository as data.csv.

8. METHODOLOGY

The project follows the following steps:

Step 1: Collect Dataset

Student data containing study hours, previous marks, and final marks is stored in a CSV file.

Step 2: Load Dataset

The Pandas library is used to read the CSV file.

Step 3: Select Features

The following two features are selected:

- Study Hours
- Previous Marks

Step 4: Select Target

The target value is:

- Final Marks

Step 5: Split Dataset

The dataset is divided into:

- Training data
- Testing data

In this project, 80% of the data is used for training and 20% is used for testing.

Step 6: Train the Model

The Linear Regression algorithm is trained using the training data.

Step 7: Evaluate the Model

The trained model is tested using the testing data. The R^2 score is calculated to evaluate the model.

Step 8: Make Prediction

The user enters study hours and previous marks. The trained model predicts the final marks.

9. LINEAR REGRESSION

Linear Regression is a supervised Machine Learning algorithm used to predict a continuous numerical value.

In this project, the model learns the relationship between:

Study Hours + Previous Marks → Final Marks

The general form of a linear regression equation is:

$$y = b_0 + b_1x_1 + b_2x_2$$

Where:

- y = predicted final marks
- b_0 = intercept
- x_1 = study hours
- x_2 = previous marks
- b_1 and b_2 = model coefficients

The model learns these values from the training dataset.

10. TRAINING AND TESTING

The dataset is divided into two parts.

Training Data

Training data is used by the Machine Learning model to learn the relationship between the input features and final marks.

Testing Data

Testing data is used to check how well the trained model performs on data that was not used during training.

The project uses an 80:20 split.

80% → Training Data

20% → Testing Data

The `train_test_split()` function from Scikit-learn is used for this purpose.

11. MODEL EVALUATION

The model is evaluated using the **R^2 score**.

R^2 , also called the coefficient of determination, measures how well the model explains the variation in the target value.

An R^2 score closer to 1 indicates that the model fits the given data well.

The program displays the R^2 score after testing the model.

Actual Project Result

Run the project using:

```
python main.py
```

The program displays:

Model R2 Score: _____

Enter the exact R^2 score displayed by your program here.

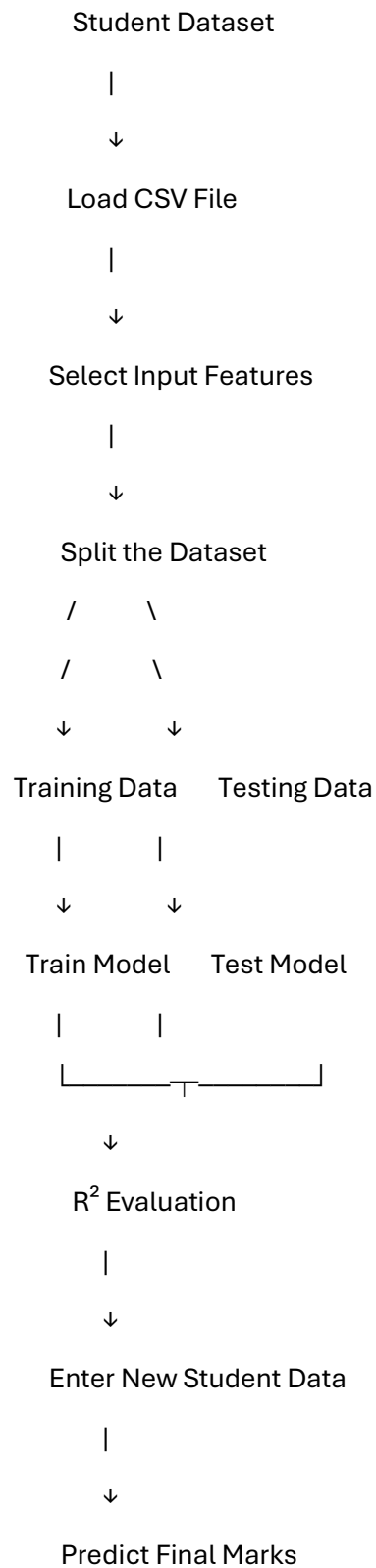
For example:

R^2 Score = 0.XX

Do not put 99% unless that is actually what your program displays.

12. SYSTEM WORKFLOW

The working of the project can be represented as:



13. IMPLEMENTATION

The main Python program is stored in main.py.

The program performs the following operations:

1. Imports the required libraries.
2. Reads the dataset using Pandas.
3. Selects the input features.
4. Selects the target value.
5. Splits the dataset into training and testing data.
6. Creates a Linear Regression model.
7. Trains the model.
8. Calculates the R^2 score.
9. Takes student information from the user.
10. Predicts the final marks.

14. SAMPLE INPUT AND OUTPUT

Input

```
=====
STUDENT MARKS PREDICTION
=====
```

Model R2 Score: 0.XX

Model Accuracy: XX.XX %

Enter study hours: 6

Enter previous marks: 75

Output

```
-----
Predicted Final Marks: XX.XX
-----
```

Prediction completed successfully!

Note: Replace XX.XX with the actual values displayed when you run your program.

15. PROJECT STRUCTURE

The project contains the following files:

student-marks-prediction/

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├─ main.py

├─ data.csv

├─ requirements.txt

└─ README.md

main.py

Contains the Python code for loading the dataset, training the model, evaluating it, and predicting final marks.

data.csv

Contains the student dataset used by the Machine Learning model.

requirements.txt

Contains the Python libraries required to run the project.

README.md

Contains project information and instructions for installation and execution.

16. INSTALLATION AND EXECUTION

The project can be executed through the command line.

Step 1: Clone the Repository

git clone <https://github.com/NarendraMewada25/student-marks-prediction.git>

Step 2: Open the Project Folder

cd student-marks-prediction

Step 3: Install Dependencies

pip install -r requirements.txt

Step 4: Run the Program

python main.py

Step 5: Enter Student Details

The program asks the user to enter:

- Study hours
- Previous marks

The program then displays the predicted final marks.

17. ADVANTAGES

The main advantages of the project are:

1. Simple and easy to use.
 2. Easy to understand and implement.
 3. Uses a basic Machine Learning algorithm.
 4. Runs directly from the command line.
 5. Requires only a small dataset.
 6. Provides quick predictions.
 7. Demonstrates the basic Machine Learning workflow.
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18. LIMITATIONS

The current project has some limitations:

1. The dataset is small.
2. Only two input features are used.
3. The prediction depends on the quality of the dataset.
4. Real student performance can depend on many other factors.
5. The model may not provide accurate predictions for students whose characteristics are very different from the training data.

Therefore, the result should be considered an estimated prediction rather than an exact final mark.

19. FUTURE SCOPE

The project can be improved in several ways in the future.

Possible improvements include:

1. Using a larger real-world dataset.
2. Adding attendance as an input feature.
3. Adding assignment and internal examination marks.
4. Comparing multiple Machine Learning algorithms.
5. Improving model evaluation.
6. Creating graphs to visualize student performance.

7. Developing a web-based interface.
8. Storing student records in a database.

These improvements can make the system more useful for larger-scale student performance analysis.

20. CONCLUSION

The Student Marks Prediction project demonstrates how Machine Learning can be used to predict a student's final marks.

The project uses Study Hours and Previous Marks as input features and Final Marks as the target value. A Linear Regression model is trained using the student dataset. The dataset is divided into training and testing data, and the model is evaluated using the R^2 score.

The project provides a simple example of the complete Machine Learning process, starting from data loading and preprocessing to model training, evaluation, and prediction.

Overall, the project helped in understanding the basic implementation of supervised Machine Learning using Python and Scikit-learn.

21. REFERENCES

1. Python Documentation
 2. Pandas Documentation
 3. Scikit-learn Documentation
 4. Machine Learning concepts related to Linear Regression
 5. Course material for Fundamentals of Artificial Intelligence and Machine Learning
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22. GITHUB REPOSITORY

GitHub Repository:

<https://github.com/NarendraMewada25/student-marks-prediction>

Repository Visibility: Public

Main Branch: main

FINAL WORD FORMATTING

In Word, I recommend:

- **Font:** Times New Roman
- **Body:** 12 pt

- **Main headings:** 14 pt, Bold
- **Title:** 18 to 20 pt, Bold
- **Line spacing:** 1.5
- **Alignment:** Justified
- **Page numbers:** Bottom center
- Put the **title page separately**
- Add a **Table of Contents** after the title page